



Multichannel Motion Artefact Correction in fNIRS

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Note to my Supervisor and Inspector

This document is unfortunately heavily condensed. For a full detailed report including figures and tables, I would strongly encourage (and appreciate) that you read the full report in the project repository

Abstract

Abstract can be found in the **full report**.

1 Introduction

Functional near-infrared spectroscopy (fNIRS) is a rapidly expanding non-invasive brain-imaging technique that uses near-infrared light to monitor haemodynamic changes in the outer layers of the brain. When neurons in the brain are active, their metabolic demand increases and so, to ensure they are not deprived of oxygen or glucose, local blood vessels will start to dilate in response to the increased activity. Due to the difference in absorption spectra of oxygenated and deoxygenated blood (oxyhaemoglobin (HbO) and deoxyhaemoglobin (HbR)), information about the dilation of these blood vessels can be inferred through a change in the optical density received in the source/detector pairs. These measurements of localised changes provide a proxy measurement of brain activity in their corresponding regions.

fNIRS as a brain imaging modality has grown in popularity due to it being highly-portable (especially compared to other modalities such as fMRI), non-invasive, lower cost, and the fact that it can be used during physical activity, a quality not common among brain imaging modalities. These advantages come with the practical drawback of fNIRS signals often being contaminated by physiological noise, motion artefacts, and setup-related noise, all of which can distort the measured signal and make interpretation more difficult.

In fNIRS, optodes are placed on the scalp of the participant, making it less sensitive to motion artefacts arising from head and scalp movement than other brain imaging modalities [1]. Despite this, motion artefacts are still an extremely prevalent problem in fNIRS literature. Researchers have reported that the artefacts caused by subjects' movements can significantly influence the quality of fNIRS signals [2] by reducing the signal-to-noise ratio (SNR) [3]. Researchers have also shown that using MA correction techniques can reduce this negative effect, increasing classification accuracy in cognition experiments. [4]

Many different approaches to motion artefact correction have been proposed for fNIRS. Huang et al. [4] provides a broad review of techniques developed up to that point, outlining their origins, operating principles, implementation considerations, and underlying assumptions. A number of further studies have also compared the performance of these methods under different experimental settings. However, despite the variety of available approaches, the literature remains difficult to reproduce consistently in practice, as important implementation details and processing choices are often under-specified. This makes it difficult both to evaluate methods fairly and to assess how well they generalise to newer fNIRS systems. For this reason, part of this project involved implementing the relevant processing and correction methods directly in Python, such that each stage of the pipeline could be made explicit and controlled.

This project therefore investigates a cluster-based motion correction method for high-density fNIRS data. The main technical contribution is a new MA correction approach that combines Global Variance of the Temporal Derivative (GVTD) [5] with targeted Principal Component Analysis (tPCA) [6]. Channels are grouped into local clusters using a spatially motivated clustering rule based on source-detector locations and separations. GVTD is then used to identify motion-contaminated periods within each cluster, after which tPCA is applied as a targeted correction step.

To support this investigation, a custom task-specific Python pipeline was developed to process SNIRF recordings, reimplement published correction methods transparently, and run preprocessing and evaluation on experimental data. Baseline approaches including no correction and Temporal Derivative Distribution Repair (TDDR) were also implemented for comparison. TDDR was chosen as the main baseline because, like GVTD, it operates in the temporal-derivative domain, but unlike GVTD it is applied independently to each channel. This made it a useful point of comparison between single-channel and multichannel correction strategies.

Overall, the main outcome of this project is the implementation of a cluster-based GVTD-gated tPCA correction method together with a Python pipeline for loading SNIRF recordings, preprocessing signals, applying motion correction, and evaluating outputs.

Experimental recordings were collected under finger-tapping conditions with varying levels of imposed head motion, with the aim of assessing whether correction could recover more plausible task-related responses and reduce the impact of motion contamination.

Although limitations in the quality and quantity of the collected data restrict the strength of the final empirical conclusions, the project provides a full implementation of the proposed method and a structured experimental pipeline for investigating multichannel correction in dense fNIRS systems. The remainder of this report reviews the relevant literature, describes the proposed method and implementation, and discusses the extent to which the resulting experiments support its use.

2 Background and Related Work

2.1 Motion Artefacts in fNIRS

Motion artefacts in fNIRS arise when subject movement alters optode–scalp coupling, producing signal changes unrelated to the underlying haemodynamic response [6]. In higher-density fNIRS systems, these effects are often not isolated to a single measurement, but can affect multiple nearby channels at the same time [5].

In the recorded time series, motion artefacts are most commonly described as *spikes* and *baseline shifts*. Spikes appear as brief and abrupt fluctuations, while baseline shifts appear as step-like changes that displace the signal to a new level.

Some artefacts also appear in a slower drift-like form, which can be more difficult to distinguish from genuine haemodynamic trends. These slower distortions are not the primary focus of the correction methods considered in this project.

2.2 Existing Approaches to Motion Artefact Correction

A wide range of motion artefact correction methods have been proposed for fNIRS. Huang et al. [4] group these into hardware-based solutions, which use additional auxiliary measurements (for example an accelerometer), and signal-processing-based solutions, which act directly from the recorded signals. In this project only signal-processing-based solutions are considered, as the aim is to develop an algorithm that operates directly on the recorded signal time series.

Within the signal-processing-based literature, methods mainly differ in: (i) their assumptions about artefact structure; (ii) whether they operate on channels independently (single-channel) or exploit relationships across multiple channels (multichannel); (iii) whether they require a motion detection step as a pre-requisite, and (iv) in which step of the preprocessing pipeline they are used. The vast majority of proposed methods are single-channel as it was only recently that high-density fNIRS systems were standardised, with the first commercially available HD system being the Gowerlabs LUMO released in 2019 [7]. Until then high-density systems were typically custom-built lab prototypes. Additionally, the majority of correction methods act upon the Optical Density (OD) signal, rather than the raw light intensity signal, and some even work directly on the HbO and HbR concentration signals themselves.

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2.3 Selected Approaches

The following subsection has been **heavily** thinned down due to space constraints. For full details, please refer to the corresponding section in the **full report**.

tPCA Targeted Principal Component Analysis (tPCA) is a multichannel motion artefact correction method that decomposes shared spatial covariation across channels during motion-contaminated periods to identify and remove artefact components [6]. It is a gated approach, meaning it requires a prior motion detection step to identify contaminated epochs before correction can be applied. tPCA has been shown to outperform wavelet-based and spline-based approaches in various fNIRS setups [4], though its effectiveness is highly dependent on the quality of this preceding detection step [6]. Notably, the motion detection methods used before tPCA in existing literature are almost exclusively single-channel, and multichannel alternatives are rarely if ever explored — a key motivation for this project.

GVTD The Global Variance of the Temporal Derivative (GVTD) [5] is a multichannel motion detection algorithm that computes the root-mean-square of the temporal derivative across all channels at each time point, producing a single time-series that reflects global signal instability. When combined with a threshold, it generates a channel-wide motion-identifying mask. GVTD was shown to correlate strongly with external motion measures and to achieve high sensitivity and specificity in its introductory paper [5]. While it has appeared in several studies [5, 4, 8, 9], it has predominantly been used for trial rejection and censoring rather than as a detection step for correction

algorithms. This, combined with the absence of a multichannel detection stage for tPCA, provided strong motivation for evaluating their combination — which, to the best of my knowledge, has not been explicitly explored in prior work.

TDDR TDDR is a single-channel MA correction method [1]. TDDR was chosen as the main baseline for three reasons: First, it has been reported to perform strongly on abrupt artefacts such as spikes and baseline shifts, which are also the main artefact types targeted by GVTd and tPCA. Second, like GVTd, it operates in the temporal-derivative domain, making it a useful comparison point between single-channel and multichannel derivative-based approaches. Third, it is widely regarded in recent fNIRS literature as an effective MA correction method.

2.4 Summary

Broadly, comparative studies reviewed by Huang et al. [4] suggest that no single correction method is universally best. Various methods have shown good performance under different conditions, with results depending on the dataset, artefact type, and evaluation metric used.

3 Data and Experimental Setup

The following section has been **heavily** thinned down due to space constraints. For full details, please refer to the corresponding section in the **full report**.

As part of the evaluation of my proposed method, and as a means of empirically evaluating my processing pipeline, a part of this project was dedicated to data collection. The purpose of the recordings is to produce a predictable motor-task response under varying degrees of instructed motion, ranging from no motion to substantial head motion.

Recordings were collected from 5 participants using the Gowerlabs LUMO headset under instructed conditions. The following sections describe the hardware, recording protocol, the dataset collected, and considerations after a review of the dataset.

Unfortunately the recordings taken were all of very low quality. This has had a large impact on the entire evaluation portion of this project, removing the ability to use standard evaluation techniques. This is discussed in more detail at the end of this section.

3.1 Gowerlabs LUMO System

The hardware used for both data collection and method development was the Gowerlabs LUMO headset, a high-density fNIRS system consisting of a flexible cap populated by hexagonal nodes, each containing 3 sources and 4 detectors. Measurements were acquired at two wavelengths (735nm and 850nm) and sampled at 6.25Hz. The nodal structure of the headset is particularly relevant to this work, as it naturally defines local clusters of channels, creating scope to investigate whether motion artefacts can be corrected more effectively by exploiting spatial locality rather than treating channels independently.

3.2 Recording Protocol

Task Selection For this project, a motor task, right hand finger tapping, was selected because it evokes a predictable and reproducible response in a single superficial cortical region that is well suited to fNIRS measurement, while also being straightforward to explain. Compared to cognitive tasks, finger tapping is easier to perform consistently while varying the degree of instructed head motion.

Motion Conditions Each recording contained three trials with increasing instructed motion: NHM, SHM, and LHM, adapted from Al-Omairi et al. [10]. NHM contained no instructed motion, while SHM and LHM each included 1,s of motion per 30,s task-rest block to keep artefacts infrequent for TDDR and GVTd [1, 5]. SHM used facial movements shown to induce LUMO artefacts [11], while LHM used rapid head shaking. The trials were ordered from least to most motion to reduce early source-detector decoupling.

Block Design Each recording consisted of three identical trials separated by 20,s baselines. Each trial contained eight 30,s task-rest blocks, with 10,s of finger tapping followed by 20,s of rest, consistent with prior motor-task fNIRS studies [10]. Visual cues were used without countdowns to avoid preparatory responses before task onset [12]. All event timings were recorded as sample markers for later analysis.

3.3 Recording Procedure

The protocol instructions were displayed on a standard Windows laptop running a custom Python script, which used PsychoPy for display design and timing and PyLSL for accurate time-marker transmission. This script would display the instructions, cues and would send the time-markers to the Gowerlabs software to store alongside the generated file. The instructions, python code used, and recording details can be found in `/psychopy/` in the project repository.

3.4 Collected Dataset

Recordings were collected from 5 participants using the Gowerlabs LUMO headset across three two-hour recording sessions. Each participant completed one recording following the protocol described above. Although the full dataset was retained for inspection and preprocessing, its use for quantitative evaluation was limited by signal quality issues, which are discussed in the following subsection.

3.5 Limitations of the Data

Following collection, an initial visual inspection of the recordings suggested that the data was of acceptable quality. However, after processing the recordings through the full preprocessing pipeline and evaluating them using standard fNIRS analysis (block-averaging-based and GLM-based), the dataset did not show clear task-evoked activations at the expected times for any participant. Since this was inconsistent with the intended motor-task design, several stages of the processing code were reviewed and revised, and multiple analysis attempts were repeated to check whether the issue arose from the implementation rather than the recordings themselves. As a result, the collected dataset was not considered sufficiently reliable for drawing strong conclusions about task-evoked activation recovery. It was still useful for testing and developing the processing pipeline, and parts of it will still be used in the evaluation, but its limitations should be recognised when interpreting the confidence in the evaluation results presented later in this report.

4 Methodology

The following section has been **heavily** thinned down due to space constraints. For full details, please refer to the corresponding section in the **full report**.

4.1 Overview of the Processing Pipeline

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4.2 Signal Preparation

4.2.1 Channel Selection

Many of the 3456 channels are dropped prior to any processing.

Two subsets of channels are dropped before further processing: one because their nodes were not considered for analysis, and the other because their source-detector separations were either too short or too long to capture meaningful signal. In adult fNIRS channel distances can range from ≈ 8 mm to ≈ 50 mm, usually grouped into short-separation (8 – 15 mm) and long separation (20 – 50 mm).[13, 14, 5]. To exclude signals in the lower SNR range, the bounds 10 mm and 40 mm were chosen.

Performing this channel pruning on LUMOs 3456 channels, keeping only 6 of 12 nodes, and using $min = 10$ mm, $max = 40$ mm, reduces the number of channels to just 344.

4.2.2 Conversion to Optical Density

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4.2.3 Signal Quality Evaluation

Additionally, before any correction is applied, signal quality is assessed in order to identify poor channels and to provide an indication of the overall recording quality. In the initial implementation, the intention was to discard all channels falling below a chosen signal-quality threshold. However, because the collected recordings were unusually noisy, this would have resulted in the removal of too many

channels, potentially undermining the effectiveness of the multichannel correction methods. For this reason, the approach was changed so that signal quality was tracked and retained as a reference measure, rather than being used as a strict pruning criterion.

The signal quality metric used for this was the Scalp Coupling Index (SCI), computed on the baseline period, implemented following the PHOEBE paper [15]. SCI estimates channel quality by isolating the cardiac-frequency component of the raw intensity signals and measuring the correlation between the two wavelengths, with stronger agreement indicating better optical coupling and higher signal quality. Other signal quality metrics exist, including Peak Spectral Power (PSP), signal-to-noise ratio (SNR), signal quality index (SQI) and coefficient of variation (CV), but these were not used in this project. [15, 16]

4.3 Correction Pipelines

After signal preparation, the OD channels were passed through each of the correction pipelines considered in this project: **No Correction**, used as a baseline; **TDDR**; the proposed cluster-based GVTG-gated tPCA method, referred to as **GGtPCA**; and a variant of this method incorporating the GVTG channel filter, referred to as **fGGtPCA**.

4.3.1 No Correction

In this pipeline, no motion artefact correction is applied. The input OD signal is passed directly to the output, providing a reference for later analysis of the impact of the other correction pipelines.

4.3.2 TDDR

This subsection is omitted from the condensed version of the report due to space constraints. Please refer to the corresponding section in the full report.

4.3.3 Cluster-Based GVTG-Gated tPCA Method (GGtPCA) Overview

The proposed multichannel correction pipeline is referred to as **GGtPCA**. In this pipeline, the input OD signal is first split into local channel clusters, after which GVTG is used within each cluster to identify motion-contaminated periods. These periods are then converted into correction bins, and tPCA is applied only to the affected parts of each cluster signal.

A variant of this method, **fGGtPCA**, additionally enables the optional GVTG channel filter before motion detection and correction. In this variant, GVTG bin calculation and tPCA correction are performed only on channels that pass the filter, after which all channels, including those identified as noisy, are concatenated back together to form the final output.

4.4 Detailed Structure of GGtPCA

4.4.1 Clustering

The purpose of clustering in this method is to allow motion detection and correction to be performed on local groups of channels rather than across the full recording at once. In a high-density system, nearby channels may share motion artefacts because they are affected by the same local optode movement, while more distant channels may be less related. Defining local clusters therefore provides a way to retain the multichannel nature of the correction step, while reducing the risk of combining unrelated channels into a single global correction.

Additionally, channels with different source-detector separations are not structurally equivalent, since their depth sensitivity differs and they can contain different proportions of superficial and cortical signal. Grouping channels with very different separations into the same multichannel correction step therefore risks combining signals with different underlying structure, which could cause issues during tPCA. Because of this, when defining clusters, the range of source-detector separations must be considered.

The goal of clustering, therefore, is to divide the full set of channels into distinct subsets such that each channel belongs to exactly one subset, with channels within each subset being spatially and structurally related. Two clustering methods are proposed in this project:

Node-Pair Clustering. The node-pair clustering algorithm forms clusters according to the pair of headset nodes to which a channel belongs. Under this scheme, a channel between nodes i and j is placed in the same cluster as a channel between j and i , so each cluster corresponds to a unique pair of hexagons on the LUMO headset. Given the 3-source, 4-detector configuration of a LUMO node, with 2 wavelengths measured for each source-detector pairing, this gives a maximum of $3 * 4 * 2 * 2 = 48$ channels in clusters where $i \neq j$ and $3 * 4 * 2 = 24$ channels where $i = j$.

The motivation for this approach is that motion artefacts in a channel are likely to originate from movement at one of its two contributing nodes, meaning that channels sharing the same node pair are likely to be affected by similar local motion. This produces spatially meaningful clusters that are directly tied to the physical headset structure.

A limitation of this clustering method is that the effective number of channels per cluster can become fairly small in practice. Although the maximum cluster sizes of 24 and 48 channels are reasonable, this number is reduced after excluding noisy channels and removing channels with overly short or long source-detector separations, leaving fewer channels for the later GVTd and tPCA stages.

K-Means Clustering. This paragraph is omitted from the condensed version of the report due to space constraints. Please refer to the corresponding section in the full report.

4.4.2 GVTd-Based Motion Detection

For each cluster, GVTd-based motion detection was performed using the `gvtd.pipeline`. In the fGGtPCA variant, an initial GVTd calculation was first performed on the unfiltered cluster signal in order to apply the GVTd channel filter and identify noisy channels. While the original GVTd implementation uses a 60 s window to consider noisy channels, this implementation uses a window of 20 s to better line up with the block design used, as one can expect there to be multiple motion-free 20 s periods but few 60 s ones.

After optionally performing a channel filter, the main GVTd calculation was then performed on a temporally filtered and trimmed version of the retained cluster OD signal, using its own high-pass and low-pass cutoffs chosen specifically for motion detection rather than later signal processing. GVTd was then thresholded using the standard threshold rule with the tunable constant c . If no valid channels or no motion periods were found, the cluster was passed through unchanged.

Although the GVTd-tPCA process is iterated multiple times, in fGGtPCA the noisy channel mask is computed only once, before the first iteration, and is kept the same through each iteration.

4.4.3 Motion Bin Construction

The thresholded GVTd output was converted into a binary motion mask, from which contiguous motion bins were constructed. Each detected period was padded on both sides using a configurable temporal padding value so that the full motion-contaminated region, rather than only the threshold crossing itself, could be corrected. Since GVTd was computed on the filtered and trimmed signal, the resulting bins were then mapped back to the indexing of the original cluster signal before correction was applied.

4.4.4 Targeted PCA Correction

If a cluster contained at least one valid channel and at least one motion bin, tPCA was applied to the selected channels within that cluster. Only the samples falling inside the detected motion bins were used in this step, so that correction was targeted specifically at the motion-contaminated periods rather than the full signal. The binned segments were concatenated in time, PCA was applied to this concatenated signal, and the leading components were removed according to the configured variance target and maximum number of principal components.

After tPCA correction, the corrected bins were written back into their original temporal positions in the cluster signal. This reconstruction step follows the same general idea used in standard spline interpolation and later tPCA implementations, where corrected motion segments are shifted back into the full signal in a way that preserves continuity at the bin boundaries [17, 18, 6]. In the implementation used here, this was done by aligning each corrected bin to the mean signal immediately before and after the bin using a short alignment window, and then applying a small taper at both bin edges to smooth the transition back into the uncorrected signal. Channels not selected for correction were left unchanged.

4.4.5 Reconstruction of the Corrected Signal

For each cluster, only the channels selected for correction were updated, while all other channels were left unchanged. After all clusters had been processed independently, their corrected signals and metadata were concatenated back together to form the final corrected OD output returned by the pipeline.

4.5 Post-Correction Processing

After MA correction there are still a few steps before the data can be analysed using the various evaluation methods. These include a temporal filter, conversion from OD to HbO and HbR concentrations, and splitting the recording into the three individual trials (NHM, SHM, LHM).

4.5.1 Band-Pass Filtering

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4.5.2 Modified Beer-Lambert Law Conversion

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4.5.3 Trial Slicing

In order to compare the three trials in each recording fairly, all preprocessing is done across the entire recording rather than on individual slices. This kept a fairer comparison when looking at the impact of the motion artefact correction algorithms in the context of the entire processing pipeline. Now that the signals have been processed into their final form (HbO/HbR) they should be split in order to be evaluated independently.

The trial slicing was simple to perform. Each trial was defined to start at the beginning of its preceding baseline period and to end at the end of its following baseline period. This meant that every extracted trial contained both its own task-rest blocks and the baseline segments immediately surrounding them. Since the baseline periods between consecutive trials were shared, the extracted trial windows overlapped slightly at their boundaries. This was intentional, as it allowed each trial to retain a complete local baseline context for later evaluation.

4.6 Unused Processing Methods

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4.7 Implementation Details

The following subsection has been **heavily** thinned down due to space constraints. For full details, please refer to the corresponding section in the **full report**.

The full processing and evaluation pipeline was implemented in Python 3.11, using NumPy and SciPy for numerical computation, pandas for channel metadata management, and matplotlib for visualisation. All processing and correction methods were implemented from scratch. This choice was made to ensure full transparency over every stage of the pipeline and to allow the flexibility needed when developing a custom correction method.

SNIRF File Handling. As mentioned, recordings were saved in the LUF format and converted to SNIRF prior to processing. The `snirf_processing` module handles loading these files, extracting the raw intensity array, channel metadata, sampling rate, and stimulus markers into easier to use formats (dictionaries, NumPy arrays).

Correction Pipeline Architecture. The correction stage is built around a class hierarchy designed to make each method modular and interchangeable. A base `CorrectionPipeline` class defines a common interface: it accepts a standardised `CorrectionPipelineIn` dataclass containing the OD signal, channel metadata, sampling rate, and parameter dictionary, and returns a `CorrectionPipelineOut` dataclass with the corrected signal and associated metadata. Each concrete correction method (`NoCorrection`, `TDDR`, `GGtPCA`, `fGGtPCA`) subclasses this base and implements its own pipeline method. Adding a new correction method to the comparison therefore requires only writing a new subclass and passing it to the pipeline.

Signal Representation Conventions. This paragraph is omitted from the condensed version of the report due to space constraints. Please refer to the corresponding section in the full report.

Overall Data Flow. The end-to-end pipeline takes a single SNIRF file and a shared parameter dictionary as input to the `processing_pipeline`. From this input, the processing pipeline loads the recording, applies the selected preprocessing and correction methods, and returns a Python dictionary containing the processed data products needed for later analysis. This includes both the overall recording after processing and the individual trial recordings derived from it, with intermediate sub-processing outputs still retained, such as OD signals, filtered signals, channel-filtered metadata, and final HbO/HbR concentration signals. This processed dictionary is then used by the evaluator, which applies the chosen evaluation methods using the stored outputs. The evaluator returns a second dictionary containing the resulting evaluation summaries, metrics, and any supporting method-specific outputs for every combination of trials and correction pipelines. The same parameter dictionary is passed throughout this process, so that processing, correction, and evaluation all operate under a single explicit set of settings.

Verification of Reimplemented Methods. This paragraph is omitted from the condensed version of the report due to space constraints. Please refer to the corresponding section in the full report.

4.8 Default Parameters

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5 Evaluation

5.1 Intended Evaluation Methods

The primary goal of the evaluation was to determine whether MA correction could recover a plausible task-evoked haemodynamic response from recordings contaminated by instructed head motion. Two standard fNIRS evaluation approaches were implemented for this purpose: block-averaged responses and a General Linear Model (GLM). Both methods rely on the assumption that the underlying data contains a recoverable task-related signal, and in both cases the expected result would be a characteristic increase in HbO concentration and a corresponding decrease in HbR concentration in channels over the motor cortex during the finger-tapping task periods.

The rest of this subsection is omitted from the condensed version of the report due to space constraints. Please refer to the corresponding section in the full report.

5.2 Data Quality and Its Impact

5.2.1 Signal Quality Assessment

Signal quality was assessed using the Scalp Coupling Index (SCI) computed over the initial baseline period of each recording. SCI measures the correlation between the two wavelengths at the cardiac frequency. In the PHOEBE implementation, a score above 0.8 indicates a clean channel [15], but given an initial inspection showed poor channel quality, this threshold was lowered to 0.5 in this project, hoping to increase the number of viable channels.

Four of the five participants have effectively no usable channels under this threshold, with three below 1.5%. Unfortunately, Subject 1 is the only participant with a meaningful proportion of clean channels at 20.9%, and is therefore the sole participant considered in the evaluation that follows.

5.2.2 Failure of Standard Evaluation Approaches

This subsection is omitted from the condensed version of the report due to space constraints. Please refer to the corresponding section in the full report.

5.3 Alternative Evaluation Approach

Because the dataset was not adequate for validating recovery of true task-evoked haemodynamics, the final evaluation focuses on whether the methods reduce motion-like corruption while preserving plausible signal structure, rather than whether they recover statistically convincing motor activation.

This is a significant limitation. Task-evoked recovery is the standard basis for evaluating correction methods, and without it the strongest form of validation is unavailable. However, the core contribution of this project is the design and implementation of a cluster-based correction method, and whether a method reduces motion-related corruption can still be assessed to some extent without a known underlying response [8, 4], even if the conclusions that can be drawn are necessarily weaker.

Additionally, because the evaluation is no longer bound to the motor cortex, the node restriction applied during pipeline design (Section 4.2.1) is no longer necessary. For the analysis that follows, all 12 nodes were included, expanding the channel set from 344 to 744 after distance-based pruning, increasing the statistical power for the metrics used below. Given this new larger set of channels, the channels signal quality was re-evaluated in, showing, once again, that Subject 1's recording is the only one suitable for evaluation.

The following evaluation uses three metrics, each targeting a different aspect of correction quality. The first assesses whether correction suppresses motion-related contamination during instructed motion periods. The second assesses whether correction preserves the expected anti-correlated relationship between HbO and HbR, a basic physiological property of clean fNIRS signals [19]. The third checks whether correction introduces unnecessary distortion to periods where no motion was instructed. Together, these cover artefact removal, signal preservation, and over-correction.

Each method described in this subsection has been heavily condensed in this version of the report. Please refer to the corresponding section in the full report.

5.3.1 Artefact Suppression: Event-Locked GVT D Attenuation

This metric quantifies how effectively each correction pipeline suppresses motion-related disturbances during the SHM and LHM trials. For each instructed motion event, a local baseline is estimated from the GVT D trace in the seconds before onset, and the integrated excess GVT D above this baseline is computed over the five seconds following onset. These per-event scores are aggregated using the median across the eight events in each trial, and the result is expressed as a percentage attenuation relative to the uncorrected pipeline — where 0% indicates no improvement and 100% would indicate complete removal of the motion burden. This metric is computed only for the SHM and LHM trials, since the NHM trial contains no instructed motion events. It should also be noted that the metric is not entirely neutral with respect to GGtPCA, which uses GVT D internally; the implications of this are examined in Section 5.5.4, and the remaining two evaluation metrics, which are independent of GVT D, provide a counterbalance.

5.3.2 Signal Preservation: HbO-HbR Anti-Correlation

This metric assesses whether each correction pipeline preserves the expected anti-correlated relationship between HbO and HbR, targeting signal preservation rather than artefact suppression. For each pipeline and trial, the Pearson correlation between the HbO and HbR time series is computed independently for each channel that passes the quality filter described in Section 5.2.2. This is evaluated over three temporal segments: the full trial, task periods only, and rest periods only — the latter providing a clean within-trial reference, since instructed motion was confined to task periods.

The resulting per-channel correlation values form a distribution for each pipeline, trial, and segment combination, which can be compared using summary statistics and visual inspection. A successful correction method would be expected to shift the distribution towards stronger negative correlations in the SHM and LHM task periods [19], where motion contamination degrades the relationship, while leaving the NHM and rest-period distributions largely unchanged. Although this metric cannot distinguish genuine physiological anti-correlation from other sources of shared variance, the ambiguity applies equally across all pipelines, so relative comparisons remain informative.

5.3.3 Over-Correction: Rest-Period Signal Variability

This metric assesses whether each correction pipeline over-corrects periods of the signal where no correction should be needed. Neither of the preceding metrics captures a method that uniformly flattens the signal: GVT D Attenuation would register such flattening as successful suppression, while HbO/HbR Anti-Correlation is scale-invariant and would not detect it at all. To address this gap, the standard deviation of HbO and HbR concentrations is computed over the concatenated rest periods for each channel, and expressed as a percentage change relative to the uncorrected pipeline. A negative value indicates reduced variability during rest; a positive value indicates increased variability.

Rest periods are the ideal window for this check because they should contain minimal motion contamination — particularly in the NHM trial, where any reduction in variability cannot be attributed to successful artefact removal and must represent over-correction. On its own this metric has limited interpretive value, since a method that does nothing would score perfectly, but combined with GVT D Attenuation it confirms whether suppression was targeted rather than indiscriminate.

5.4 Results

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5.5 Discussion

5.5.1 Overall Outcome of the Project

The project delivered what it set out to build: a modular Python pipeline implementing the full processing chain from raw SNIRF files to evaluated HbO/HbR outputs, with each processing and correction stage implemented transparently from scratch, including the proposed GGtPCA and fGGtPCA methods and the baseline TDDR method. Together with the experimental recordings and evaluation framework, this constitutes a complete and reproducible system for investigating multichannel motion correction in dense fNIRS data.

What the project could not deliver was a strong empirical validation. The data quality limitations described in Sections 3.5 and 5.2 meant that standard task-recovery evaluation was unavailable, and the proxy metrics used in its place can assess whether a corrected signal is more plausible but not whether it is more accurate. This limits how confidently differences between methods can be interpreted, and should be kept in mind throughout the discussion that follows.

Despite these limitations, the indicative results from the single usable recording are striking. fGGtPCA achieved 65.6% attenuation of the instructed motion response while leaving the clean-condition signal almost entirely unchanged. Its task-period correlation improvements were close to those of TDDR, but without TDDR's substantial cost to rest-period variability and clean-condition signal structure. The fact that a gated, cluster-local method can approach the suppression performance of a well-regarded single-channel method while preserving signal properties that TDDR measurably degrades is a promising result.

The remainder of this discussion examines what these results reveal about the design choices underlying GGtPCA, the nature of the trade-offs between the three correction approaches, and the limits on what can be inferred from the available evidence.

5.5.2 What the Results Suggest About the GGtPCA Design

This subsection is omitted from the condensed version of the report due to space constraints. Please refer to the corresponding section in the full report.

5.5.3 Trade-offs Between Correction Approaches

This subsection has been **heavily** thinned down in condensed version of the report due to space constraints. Please refer to the corresponding section in the full report.

The results show clear differences between the three correction pipelines, which can be traced to fundamental properties of each method's design.

TDDR's strong suppression and its cost to signal preservation both stem from the same mechanism. It operates independently on each channel and downweights all large temporal-derivative outliers without distinguishing their source, meaning it catches artefacts that a threshold-based detector would miss, but cannot leave clean segments of the signal intact. The NHM rest-period variability changes confirm that this effect extends beyond motion-contaminated periods.

GGtPCA sat at the opposite extreme. This is not a fundamental limitation of the method's design but rather evidence that unfiltered GGtPCA requires a level of channel quality that this dataset did not provide.

fGGtPCA occupied the middle ground, substantially weaker than TDDR in raw attenuation but much stronger in preservation behaviour. The gap likely reflects a fundamental difference between gated and ungated correction: fGGtPCA can only correct motion events that produce a clear GVTD exceedance within the cluster, so smaller artefacts that fall below the detection threshold are missed entirely, whereas TDDR operates on the full temporal derivative distribution and can downweight less prominent outliers. This distinction also maps onto a broader difference between single-channel and multichannel correction. TDDR treats each channel independently, whereas fGGtPCA exploits the spatial structure of high-density systems through clustering, multichannel detection, and multichannel correction. The existing tPCA literature has noted that effectiveness depends heavily on the motion detection step [6, 17], which is typically single-channel. The results presented here, while only indicative, suggest that a multichannel detection step can capture instructed motion events effectively.

Huang et al. [4] concluded that no single correction method is universally best. The results of this project are consistent with that conclusion, with TDDR and fGGtPCA representing genuinely different trade-offs rather than one being strictly better than the other. Which is preferable will depend on the application, the recording quality, and whether the downstream analysis is more sensitive to residual motion contamination or to signal distortion.

5.5.4 Limits of Inference and Threats to Validity

This subsection has been **heavily** thinned down in condensed version of the report due to space constraints. Please refer to the corresponding section in the full report.

The results are drawn from a single participant, and the patterns observed could be idiosyncratic to this particular recording rather than reflecting general properties of the methods. The GVTD attenuation metric is not entirely neutral with respect to fGGtPCA, which uses GVTD internally for detection, while the preservation and over-correction metrics are inherently more sensitive to TDDR's global dampening than to a gated method's localised correction. The fact that all three metrics, despite these opposing biases, support a consistent relative ordering of methods provides some reassurance that the overall pattern is not purely an artefact of metric choice.

The evaluation also compared only three correction approaches. Comparing fGGtPCA against additional methods such as wavelet-based correction, MARA, or CBSI would have strengthened the basis for any conclusions drawn. It is also worth considering whether the data quality amplified fGGtPCA's advantage, since the channel filter is most beneficial when many channels have poor coupling; however, noisy channels are common in practice and are routinely excluded before analysis, so the filter addresses a recurring problem rather than one specific to this dataset. The recording protocol introduces further caveats: the brief, infrequent instructed motion events were consistent with the assumptions underlying both GVTD and TDDR, and the fixed trial ordering may have affected the LHM

condition through cumulative changes in headset fit. A fuller discussion of these limitations, including the implications for more frequent or sustained motion settings, is provided in the main report.

5.5.5 Future Work

This subsection is omitted from the condensed version of the report due to space constraints. Please refer to the corresponding section in the full report.

6 Timeline

The first semester was spent building an understanding of fNIRS as a modality and exploring where within the field a project could contribute. Over the course of the semester, my focus narrowed towards motion artefact correction, with the spatial structure of the LUMO headset standing out as an underexplored advantage over the single-channel methods that dominate the existing literature. This period also involved deep reading of existing correction methods to inform the design of my own approach.

Development began at the start of the second semester. The specific decision to combine GVTD with tPCA in a cluster-based framework came during this period, as the overall pipeline took shape and the strengths of each method became clearer.

Data collection did not take place until February, and the data quality issues described in Section 3.5 were discovered soon after, leaving limited time to diagnose the issue, redesign the evaluation strategy, implement alternative metrics, and carry out the sensitivity analysis and final comparison.

7 Repository

The project repository is organised as follows. All Python source code lives under `src/`, with subfolders `pipelines/`, `evaluation/`, and `plots/` for their respective purposes. The correction pipelines themselves are in `pipelines/correction_pipeline/`, and the remaining processing stages are implemented as individual modules directly under `src/`.

Two notebook folders are included. `notebooks_results/` contains the final evaluation outputs: `sensitivity_analysis.ipynb` and `results_analysis.ipynb` reproduce the tables presented in this report, while `pipeline_view.ipynb` allows the full pipeline to be traced from start to finish with intermediate plots. `notebooks/` contains earlier exploratory work carried out during development, including experimentation with individual processing stages and parameters.

`psychopy/` contains the full recording protocol, including the PsychoPy script used to run the experiment, the instructions presented to participants, and a summary of the recording details. `data/` holds the SNIRF recordings, `data_augmentation/` contains the scripts used to add missing stimulus markers as described in Section 3.4, and `files/` contains the haemoglobin extinction coefficients used in the MBL conversion.

`notes/` contains informal notes collected throughout the project. Within it, `notes/plots/` holds all figures generated for the report, and `notes/information.ipynb` contains early but detailed documentation of the SNIRF file structure produced during the initial exploration phase.

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